

Designing a Reference Trajectory for Frozen Repeat Near-Equatorial Low Earth Orbits

Samuel Y. W. Low,⁽¹⁾

DSO National Laboratories, 12 Science Park Drive, Singapore 118225

Yongjun Moon,⁽²⁾

Satrec Initiative Co. Ltd. Daejeon, 34054, Republic of Korea

Wen Tao Liu⁽³⁾ and Chek-Wu Tan⁽⁴⁾

ST Engineering Satellite Systems, Singapore 56771

ABSTRACT

Numerical techniques for designing frozen repeat orbit as reference trajectories have been extensively studied for polar orbit missions. For near-equatorial orbits, larger perturbing forces are experienced at lower inclinations. The greater perturbations make such near-equatorial orbits' long-periodic behavior much more sensitive to changes in initial conditions. Thus, a more careful selection of optimization parameters and a longer convergence time is often required. This makes the numerical optimization stage for a near-equatorial orbit more "delicate". Past proven techniques involved a preliminary estimate, a first optimization for the initial osculating elements, and a second optimization to ensure end-cycle continuity. This paper first reviews these techniques specific to near-equatorial frozen repeat orbit design, evaluates them in comparison with polar orbit designs. Then, this paper counter-proposes a more efficient orbit design scheme that reduces the design process from two optimizations to only one optimization with a filtering stage. As a result, the orbit design process is much more efficient, and less prone to convergence issues, especially when addressing orbit end-cycle discontinuity.

¹ Mr. Samuel Y. W. Low is a Member of Technical Staff in DSO National Laboratories' Sensors Division, Singapore.

² Dr. Moon Yongjun is the team lead for the Micro Satellite Team in Satrec Initiative's Space Systems Division, Republic of Korea.

³ Mr. Liu Wen Tao is a Senior Systems Engineer in ST Satellite Systems Pte Ltd, Singapore.

⁴ Mr. Tan Chek-Wu is a Principal Systems Engineer in ST Satellite Systems Pte Ltd, Singapore.

Nomenclature

a	=	Orbit semi-major axis (units: m)
α	=	Alpha – angle enclosed between the fixed-frame and inertial velocity vectors
e	=	Orbit eccentricity (dimensionless)
E_C	=	Fixed frame cross track position errors (units: m)
E_R	=	Fixed frame radial position errors (units: m)
i	=	Orbit inclination angle (units: radians)
J_2	=	Constant used in computing J_2 (oblateness) perturbation = 0.00108263
J_3	=	Constant used in computing J_3 (pear-shaped) perturbation = -2.54503e-6
$\Delta\lambda$	=	Ground track shift in longitude (units: radians)
M	=	Orbit mean anomaly (units: radians)
n	=	Mean motion of the orbit (units: radians/second)
Ω	=	Orbit right ascension of the ascending node, or RAAN (units: radians)
ω	=	Orbit argument of periapsis (units: radians)
$R_\oplus$	=	Mean radius of the earth ~6371140 (units: m)
R_{REF}	=	Fixed frame position vector of a point along the reference trajectory (units: m)
$\sigma_\oplus$	=	Rotation rate of the Earth about its own axis (units: radians/second)
τ	=	Nodal period of orbit (units: seconds)
u	=	Argument of latitude, or sum of the argument of periapsis and true anomaly (units: radians)
Θ	=	Orbit longitude of ascending node (units: radians)
V_{REF}	=	Fixed frame velocity vector along the reference trajectory (units: m/second)
ΔV	=	Change in inertial velocity due to an impulsive maneuver (units: m/second)

1. Introduction

Frozen repeat orbit (FRO) reference trajectories are designed to repeat the ground track on Earth's surface after a fixed number of orbits in a repeat cycle, with minimum altitude variations when passing over the same geographical area of interest. This is done by optimizing the initial osculating orbit elements such that long-term secular variations are minimized. In the Earth-centered, Earth-fixed (ECEF) sense, this "freezes" the orbit passively in-place. Because this also preserves the precise viewing angles and geometry for remote sensing, FROs enable novel imaging applications such as change detection. Furthermore, at low inclinations, revisits are more frequent for low latitude ground stations than they are for mid-inclination orbits, and thus near-equatorial FROs are an attractive choice for future remote sensing science missions in low latitude regions like Singapore. This study examines design methods for such orbits to be used as a reference trajectory. The requirements given are that the reference trajectory must be both "smooth" and "continuous", in the sense of position and velocity vectors (ECEF), so as to satisfy sensitive instrumentation requirements and to prevent discontinuities being fed back to the orbit maintenance control law.

In literature, extensive study has been done on the development of FRO reference trajectories for Low-Earth Orbits (LEO), particularly via numerical methods. In nearly all cases, the orbit design begins with a careful selection of the initial osculating semi-major axis and eccentricity, to ensure that the spacecraft ascends at the same longitude of ascending node after each repeat period with minimal local altitude variation. Typically, first order estimates of the mean elements are analytically derived (including J_2 and J_3 effects). In [1], Vallado *et al* shares a search algorithm based on a user-defined repeat period. Thereafter, in [4], McClain introduced an analytical estimate to the frozen eccentricity that minimizes secular variations in argument of perigee, accounting for perturbations up to J_3 . This allowed for passive control of the eccentricity as the phase space of the analytically derived eccentricity vector would oscillate closely about the optimum frozen value. Rosengren then improved on McClain's idea two years later in [5], by developing an iterative algorithm that reduces the phase space variations of the derived eccentricity vector, until it reaches the optimum frozen value. Rosengren's approach successfully demonstrated its feasibility in the ERS-1 and ERS-2 mission in 1991 and 1995 respectively, as the actual trajectory maintained itself to the reference with spatial errors within a 300 ± 50 m torus, in ECEF [6].

However, one key issue remained then: that the reference trajectory, when looping back onto itself, was not continuous. In other words, existing reference trajectory design methods focused on optimizing the mean and initial osculating orbit elements, but did not address the end-cycle discontinuity between position vectors at the initial and final epoch of one full FRO cycle. Any discontinuity is undesirable as it may prompt an abrupt control response where the reference is used as the target for an orbit control system. This was addressed by D'Amico *et*

al in [7], for TerraSAR-X's reference trajectory design, by proposing the use of an impulsive “virtual maneuver” as an optimized mathematical parameter that forces the initial and final state vectors at the ascending nodes to be equal. This technique uses a virtual ΔV as a cost parameter. However, this also introduces a velocity vector discontinuity. Nevertheless, the reference trajectory modelling errors were not significantly compromised, since the impulsive virtual ΔV was small. TerraSAR-X's actual orbit spatial errors could be kept within the tight 250m tolerance band [8], as shared in 2010. While research in FRO design continues to mature, the focus remains largely on polar orbits, and recently also on asteroid mapping [10]. A near-equatorial FRO has not been proposed before, and the practical concern of generating a smooth, continuous reference trajectory for all state vectors, including velocity, has not been addressed since the time of TerraSAR-X, to the best of the authors' knowledge.

Therefore, this paper is motivated to offer two key contributions. First, it reviews previous aforementioned techniques in literature and applies them for the near-equatorial orbit. Differences in orbit design for the near-equatorial and polar orbit (i.e., convergence and optimization duration etc.) will also be discussed. While the techniques have been tested before, this will be a first-time validation for near-equatorial FROs. Second, this paper proposes a novel alternative orbit design method by introducing a novel “averaging filter” concept in addition to the numerical optimization of the initial osculating elements. This design scheme not only resolves the end-cycle discontinuities of both position and velocity vectors, but it is also more computationally efficient than the method of virtual maneuvers, because it reduces the orbit design problem from having to run two optimizations to only one. Finally, to prove the feasibility of this reference trajectory, an orbit maintenance and control simulation is performed, using the reference trajectories designed from both the method of virtual maneuvers, and the newly proposed average filtering scheme. A spatial error analysis is then performed, discussed, and compared between the two methods.

2. A Review of Present Numerical Methods for Orbit Design

This section applies the methods used in literature for FRO design at low inclinations and reviews the results. Figure 2.1 illustrates an example reference trajectory design scheme used in previous polar orbit missions.

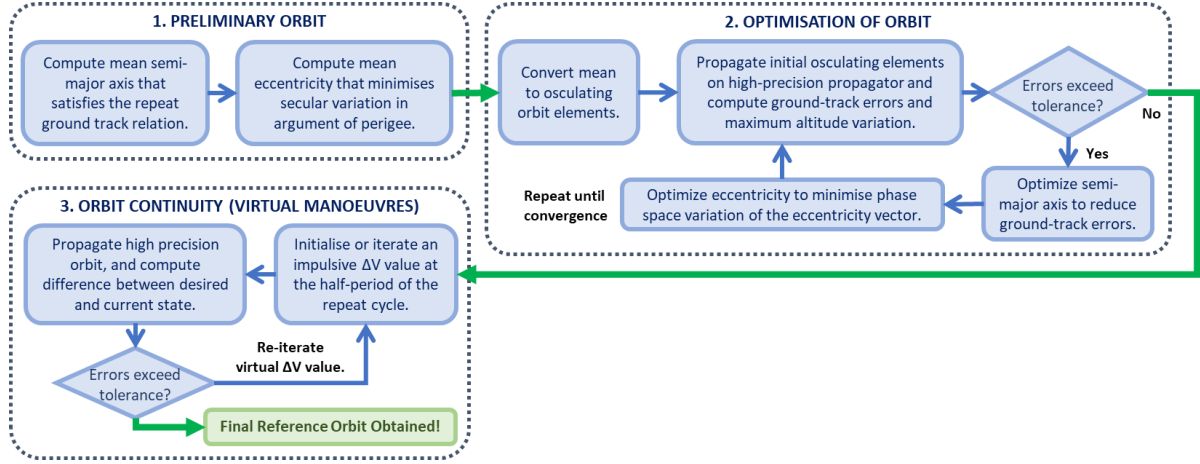


Figure 2.1 – Summary of FRO Design Scheme Used in Past Missions

For comparison, both polar and near-equatorial circular LEOs are examined, choosing a 3.9-day repeat period for purpose of example in this study. The mean semi-major axis is a parameter of the nodal period τ , and is selected such that the repeat ground track relation in (1) is satisfied. The right-side of (1) is user-defined.

$$\frac{(\sigma_{\oplus} - \dot{\Omega}) \cdot \tau}{M + \dot{\omega}} = \frac{\text{No. Sidereal Days for a Repeat Cycle}}{\text{No. Orbits in a Repeat Cycle}} \quad (1)$$

$\sigma_{\oplus}$ is the Earth's rotation rate, M is the mean anomaly, ω is the argument of perigee, and Ω is the right ascension of ascending node. Preliminary estimates to the mean semi-major axis that satisfy (1) can be found using the algorithm outlined in [1], based on a J2-only model. The initial guess mean eccentricity is computed in (2) based on [4], using the semi-major axis that satisfies (1); and ω is chosen as either 90° or 270° [4].

$$e = -\frac{J_3}{2J_2} \left(\frac{R_{\oplus}}{a(1-e^2)} \right) \sin(i) \sin(\omega) \quad (2)$$

The solutions to (1) and (2) would have provided the orbit designer with mean elements, suitable as preliminary estimates to the initial osculating semi-major axis and eccentricity. The next step to optimize them further requires a high precision orbit propagator. The optimization below uses the EGM96 gravitational model (70x70 degree and order) with sun-moon perturbations, on STK, a commercial astrodynamics software.

Numerical optimization of the initial osculating semi-major axis: Given the preliminary initial osculating semi-major axis a , and a chosen frozen repeat period P , the final longitude of ascending node at time P after the high precision orbit propagation would be Θ , computed by STK. The desired longitude of ascending node Θ_G at time P aims to be the exact same value as the longitude of ascending node in the initial epoch, as that

would be characteristic of a repeat ground track. The numerical optimization thus minimizes the loss function in (3), with $\mathbf{a}$ as the parameter, and all other initial orbit elements kept constant.

$$\Delta\Theta(\mathbf{a}) = |\Theta(\mathbf{a}) - \Theta_G| \quad (3)$$

The minimization uses Broyden's method [15] as it is more efficient than Newton-Raphson methods in numerical trajectory planning [16]. In Broyden's method, for the first iteration where $k = 0$, the first order derivative (or the Jacobian $\mathbf{J}$ for general multi-variate cases) of how the $\Theta(\mathbf{a})$ varies with $\mathbf{a}$ is numerically approximated using perturbations of the control variable $\delta\mathbf{a}$ in a Newton-Raphson fashion.

$$\mathbf{J} = \frac{\partial\Theta}{\partial\mathbf{a}} \quad (4)$$

Thereafter, when $k > 0$, there is no need to compute the Jacobian at every iteration, as Jacobian estimates follow (5), and the updated control variable follows the form of a first order update as per (6).

$$\mathbf{J}_k = \mathbf{J}_{k-1} + \frac{\Theta(\mathbf{a}_k) - \Theta(\mathbf{a}_{k-1}) - \mathbf{J}_{k-1}(\mathbf{a}_k - \mathbf{a}_{k-1})}{\|\mathbf{a}_k - \mathbf{a}_{k-1}\|^2} \quad (5)$$

$$\mathbf{a}_{k+1} = \mathbf{a}_k - \mathbf{J}_k^{-1}[\Theta(\mathbf{a}_k) - \Theta_G] \quad (6)$$

Thus, using the numerical methods in (5) and (6), the optimization problem is succinctly described as:

$$\mathbf{a}^* = \arg \min_{\mathbf{a}} (|\Theta(\mathbf{a}) - \Theta_G|) \quad (7)$$

Numerical optimization of the initial osculating eccentricity: The phase space variations of the eccentricity vector were minimized using Rosengren's algorithm [5]. The key idea is that an optimum mean frozen eccentricity vector (e_{x0}, e_{y0}) exists, and is estimated as the center of the "circle" plot of the eccentricity phase space variations (see Figure 2.2A and 2.3A). Then, there is an optimal initial osculating eccentricity, $\mathbf{e}^*$, at orbit insertion that minimizes the orbit's long-term mean eccentricity variation about this frozen value. The mean eccentricity $\mathbf{e}_{mean}$ may approach this frozen value via iterative corrections in the initial osculating eccentricity, $\mathbf{e}$, to reduce the "diameter" of the phase space variations [5]. The eccentricity vector components are defined in (8). The loss function used in (9) is the mean-squared error between the mean long eccentricity components and the optimum frozen mean eccentricity (e_{x0}, e_{y0}), for N samples through 30 ground track repeats at 30s sampling interval.

$$e_x(t, \mathbf{e}) = e_{mean}(t, \mathbf{e}) \cos \omega(t, \mathbf{e}), \quad e_y(t, \mathbf{e}) = e_{mean}(t, \mathbf{e}) \sin \omega(t, \mathbf{e}) \quad (8)$$

$$\mathbf{e}^* = \arg \min_{\mathbf{e}} \frac{1}{N} \sum_{n=1}^N \left[(e_x(t_n, \mathbf{e}) - e_{x0})^2 + (e_y(t_n, \mathbf{e}) - e_{y0})^2 \right] \quad (9)$$

The resultant convergence of the mean eccentricity variation towards the frozen value is shown in the Figures 2.2 and 2.3 below, for the respective near-equatorial and polar orbit eccentricity phase space variations. The plots before optimization used the initial eccentricity guess from (2). The plots after optimization saw the "diameter" of variation reduced more than five-fold. The convergence took thrice as many iterations in the near-

equatorial orbit as compared to the polar orbit. This was expected since perturbation forces (particularly J2 oblateness) are greater at lower latitudes than at the poles as evident in the analytical J2 model [1].

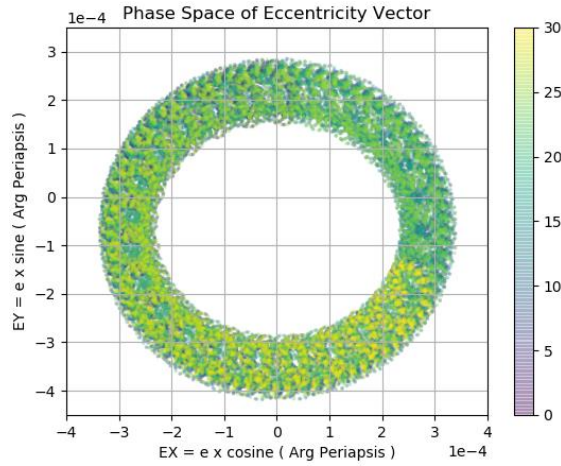


Figure 2.2A – Before Optimization for $i = 10.0^\circ$

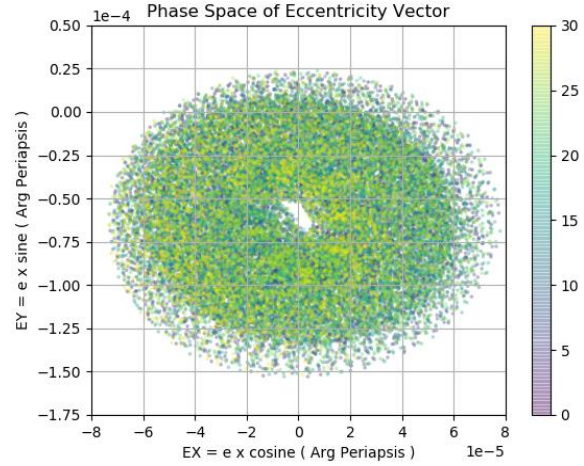


Figure 2.2B – After Optimization for $i = 10.0^\circ$

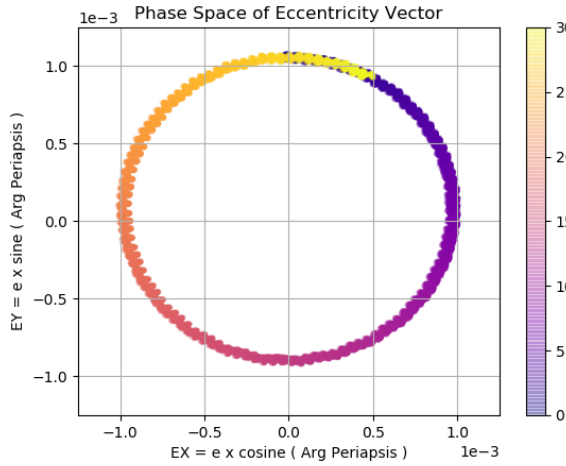


Figure 2.3A – Before Optimization for $i = 97.4^\circ$

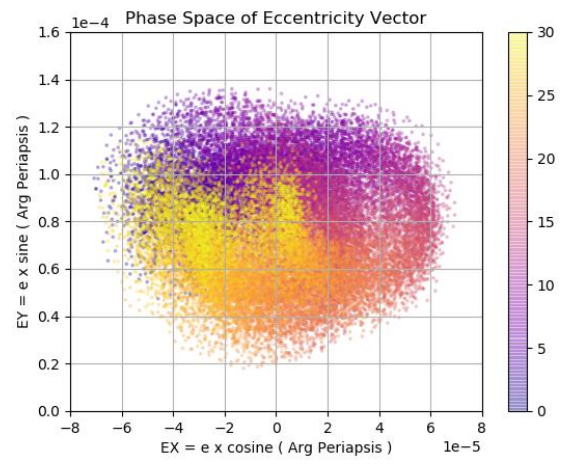


Figure 2.3B – After Optimization for $i = 97.4^\circ$

As a result of the optimizations described in (7) and (9), for a 3.9-day period FRO, the study produced the following initial osculating elements tabulated in Table 2.1 below. The orbits were then propagated for spatial error comparison on the EGM96 gravitational model, degree and order of 70, with sun and moon third body perturbations. Since the design is for the reference orbit, decay effects (drag and solar radiation) are ignored.

	Optimized Near-Equatorial Orbit		Optimized Polar Orbit	
	Mean	Osculating	Mean	Osculating
<i>Epoch</i>	1 Jan 2021 00:00:00		1 Jan 2021 00:00:00	
a^*	6934.91 km	6934.63 km	7016.79 km	7007.53 km
e^*	1.781 E-8	1.431 E-3	7.522 E-5	5.250 E-4
ω^*	90.000 ^o	90.000 ^o	270.000 ^o	270.000 ^o
i	10.000 ^o	10.000 ^o	97.440 ^o	97.440 ^o

Table 2.1 – Optimized Initial Osculating Elements

The spatial error vector is defined as the relative position vector of a spacecraft between two states with near-zero along-track error, timed one repeat cycle apart (i.e., $\mathbf{R}_{\text{TARGET}}$ minus $\mathbf{R}_{\text{REF}}$ in Figure 2.4). The spatial errors are computed every 30 seconds at checkpoints satisfying the criteria in (10). This criterion ensures that only state vectors with near-zero along-track errors are compared, which can be identified by checking if the dot product of a reference velocity vector, and the relative positions between target and reference, are zero.

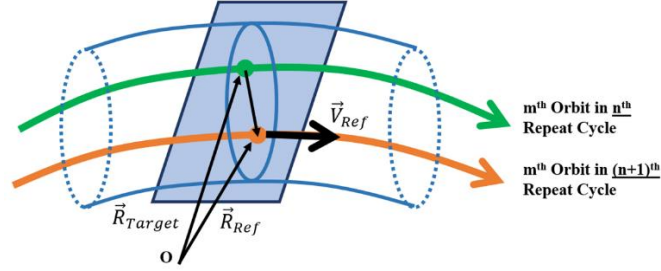


Figure 2.4 – Orbit “tube” showing the definition of and criteria for spatial error computation

$$\text{Checkpoint Criteria: } [\vec{V}_{\text{Ref}} \cdot (\vec{R}_{\text{Target}} - \vec{R}_{\text{Ref}})]_{\text{ECEF}} \approx 0 \quad (10)$$

The spatial error plots between same orbits across consecutive repeat cycles are presented in Figures 2.5 and 2.6 for the near-equatorial and polar FRO trajectories respectively, with a summary of errors in Table 2.2.

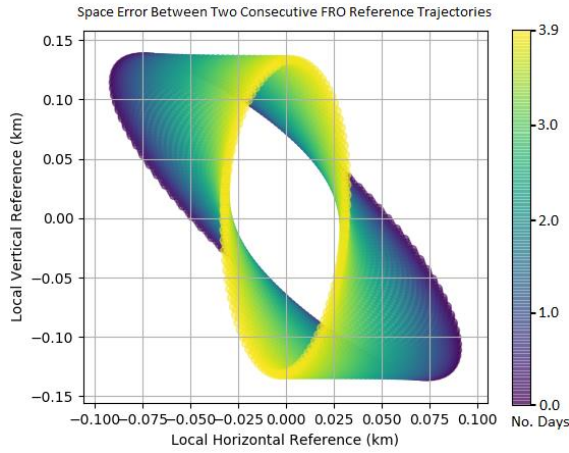


Figure 2.5 – Spatial error evolution for $i = 10^\circ$.

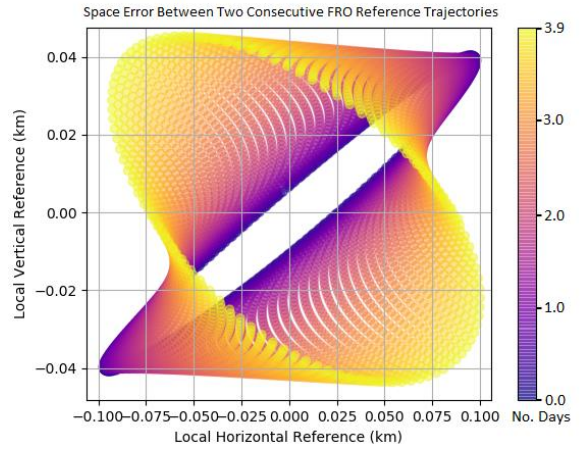


Figure 2.6 – Spatial error evolution for $i = 97.4^\circ$.

	Near-Equatorial (see Fig 2.2A)		Polar Orbit (see Fig 2.2B)	
	Mean (m)	STD (m)	Mean (m)	STD (m)
Radial	-0.012	128.473	-0.231	29.207
Off-Track	-0.677	44.447	-2.444	58.752

Table 2.2 – Summary of Spatial Errors for Both Orbit Cases

Then, the end cycle discontinuities for both orbits were computed, and tabulated in Tables 2.3A/B. Note that the position Z-coordinates are zero, as the states at the initial and final epoch are at the ascending nodes.

		Near Equatorial Orbit					
		Position Vectors			Velocity Vectors		
Cycle	Initial Epoch	X _P (km)	Y _P (km)	Z _P (km)	X _V (km/s)	Y _V (km/s)	Z _V (km/s)
1	1 Jan 2021 01:11:44	3284.632	6097.367	0.000	-6.137519	3.305119	1.319258
2	4 Jan 2021 23:00:52	3284.705	6097.469	0.000	-6.137492	3.304918	1.319337
RMS Discontinuity		0.1274275 km			0.000217649 km/s		

Table 2.3A – End-Cycle State Vector Discontinuities for the Near-Equatorial Orbit

		Polar Orbit					
		Position Vectors			Velocity Vectors		
Cycle	Initial Epoch	X _P (km)	Y _P (km)	Z _P (km)	X _V (km/s)	Y _V (km/s)	Z _V (km/s)
1	1 Jan 2021 01:13:14	-6793.145	1781.283	0.000	0.389711	1.450389	7.470276
2	4 Jan 2021 23:00:52	-6793.110	1781.373	0.000	0.389731	1.450281	7.470287
RMS Discontinuity		0.096165196 km			0.000110386 km/s		

Table 2.3B – End-Cycle State Vector Discontinuities for the Polar Orbit

To resolve the discontinuities, the method of virtual maneuvers proposed by D’Amico *et al* [7] was applied to both orbits, forcing the final state vector to match the initial. Broyden’s method [16] was applied, with the virtual maneuver $x = \Delta V$ vector as the control variable. The resultant position and velocity state $f(x)$ was computed using STK Astrogator, accounting for the impulsive virtual ΔV in each iteration. The same orbit propagator was used as in the optimization stage. The virtual maneuver was applied at the mid-point of the repeat cycle for both orbits. The desired goal for the final epoch state vector, y_G , is the initial state epoch state vector (so that they match). Broyden’s method was iterated, updating the virtual ΔV with a max step size of 1cm/s.

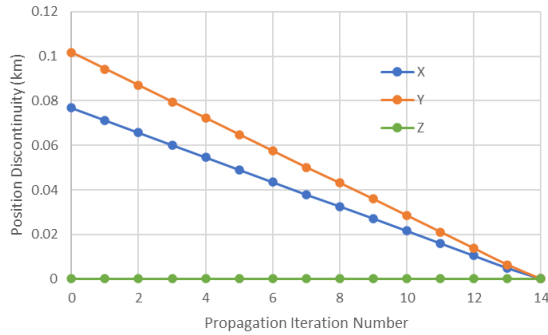


Figure 2.7 – Convergence of Position Discontinuities in the Near-Equatorial Orbit

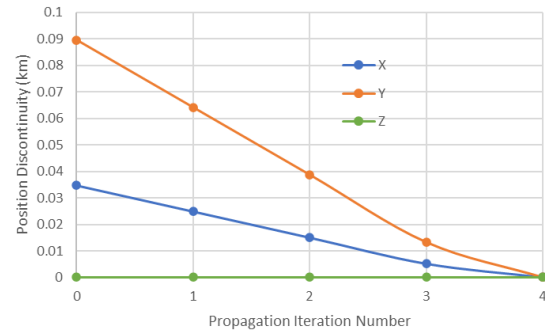


Figure 2.8 – Convergence of Position Discontinuities in the Polar Orbit

Convergence was clearly observed in Figures 2.7 and 2.8 – proving that the method of virtual maneuvers as used in TerraSAR-X [7] works on orbits with low inclinations too. Again, the Z-coordinates are zero as the end-cycle position vectors lie on the equatorial plane. The solutions to virtual maneuvers are in Table 2.4.

	ΔV_X (m/s)	ΔV_Y (m/s)	ΔV_Z (m/s)
Near-Equatorial Orbit	-4.51873E-04	-0.130608	0.137849
Polar Orbit	-9.7322E-04	-1.08071E-02	3.5213E-02

Table 2.4 – Virtual Maneuver ΔV Solutions

The study found that the optimization was longer for the near-equatorial orbit, with 15 iterations and propagations, as compared to 5 for the polar orbit. Also, both had the same maximum step sizes, but the range of permissible step sizes was almost thrice as narrow in the near-equatorial than polar orbit. A sensitivity analysis elucidates a possible reason – the precession rate of the argument of perigee, which is directly responsible for altitude variations, is far more sensitive to an impulsive thrust for near-equatorial orbits than for polar orbits. This makes convergence for the near-equatorial orbit more “delicate” to achieve. To explore why, the expression in (11), which provides the long-periodic rate of change of ω in a $J_2 + J_3$ model [1], was analyzed. All elements used in (11) to (14) were converted to mean elements.

$$\dot{\omega} = 3J_2 n \left(\frac{R_\oplus}{a} \right)^2 \left(1 - \frac{5}{4} \sin^2(i) \right) \cdot \left(1 + \frac{J_3}{2J_2} \left(\frac{R_\oplus}{a} \right) \left(\frac{\sin^2(i) - e^2 \cos^2(i)}{\sin i} \right) \frac{\sin \omega}{e} \right) \quad (11)$$

The partial derivative of (11) is found with respect to changes in the semi-major axis, and expressed and simplified with two substituted terms A and B, arranged according to J_2 and J_3 effects, for the sake of brevity:

$$\frac{\partial \dot{\omega}}{\partial a} = \frac{A(i)}{\sqrt{a^9}} + \frac{B(i, \omega)}{\sqrt{a^{11}}} \quad (12)$$

$$A(i) = -\frac{21}{2} \cdot J_2 \cdot \sqrt{GM} \cdot R_\oplus^2 \cdot \left(1 - \frac{5}{4} \sin^2(i) \right) \quad (13)$$

$$B(i, \omega) = -\frac{27}{4} \cdot J_3 \cdot \sqrt{GM} \cdot R_\oplus^3 \cdot \frac{\sin(\omega)}{e} \cdot \left(\sin(i) - \frac{5}{4} \sin^3(i) \right) \quad (14)$$

For an initial semi-major axis and velocity value, the first order change in semi-major axis due to some ΔV is derived from the classical astrodynamics’ vis-visa equation:

$$\left[\frac{\partial a}{\partial v} \right]_{a_o} \approx \frac{-2a_o}{v(a_o)} \quad (15)$$

Equation (15) was then multiplied to (12) as per chain rule to derive the first order change of the altitude variation rate ($\dot{\omega}$ -dot) with respect to an impulsive ΔV added into the system, or the sensitivity of interest. It turns out that analytically, the long-periodic $\dot{\omega}$ -dot at $i=10.0^\circ$ is about thrice as sensitive than $i=97.44^\circ$ to an input ΔV .

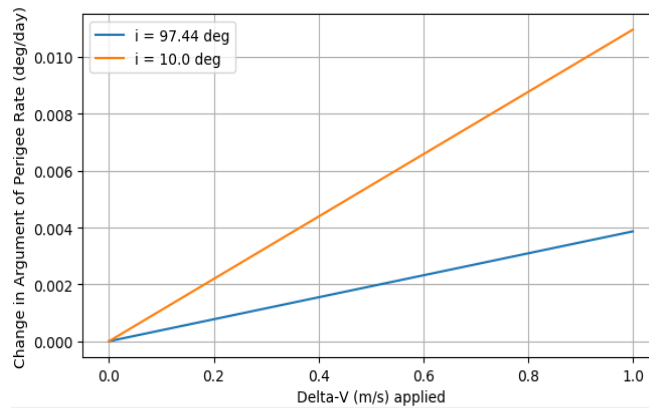


Figure 2.9 - Changes in $\dot{\omega}$ -rate (perigee precession rate), in response to an impulsive ΔV .

As a result of the increased sensitivity in altitude variations, altitude variations between consecutive reference trajectory repeat cycles were expected to increase for the near-equatorial orbit case as compared to the polar orbit. The near-equatorial orbit also needed a more “careful selection” of step sizes and other initial conditions to ensure convergence. Between consecutive repeat cycles each, this time with a virtual maneuver, the spatial errors were again computed at checkpoints with no in-track error (in similar fashion to Figures 2.2) and indeed it indicates a greater spatial error variation for the near equatorial orbit than the polar orbit. Notice two distinct spatial error evolution tracks in each plot, one before the application of the virtual ΔV , and one after.

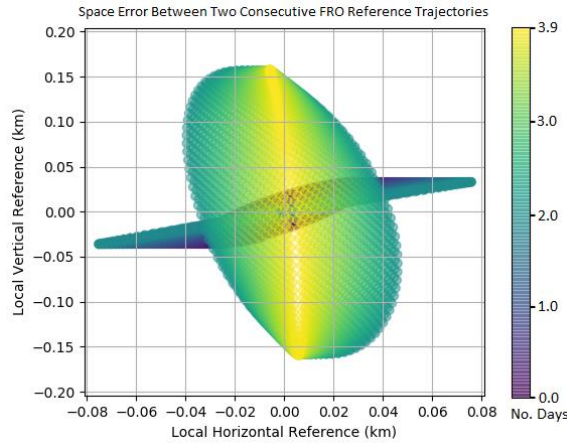


Figure 2.10 – Spatial error evolution for $i = 10^\circ$ with virtual ΔV at $t = 2.0$ days

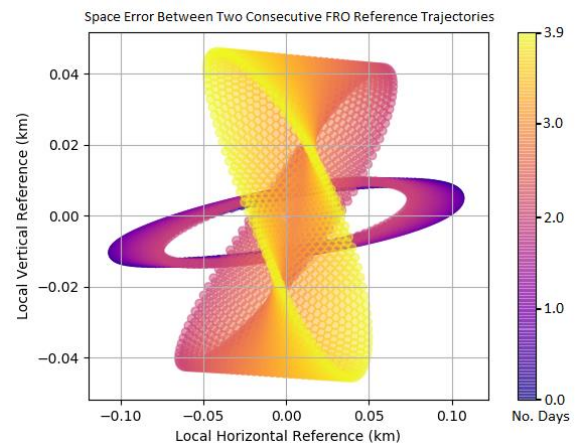


Figure 2.11 – Spatial error evolution for $i = 97.4^\circ$ with virtual ΔV at $t = 2.0$ days

In summary, this section verified that a compilation of FRO design methods in literature for polar orbits also applies to a near-equatorial orbits in designing a continuous reference trajectory with spatial errors within 10E2 meters range after a first optimization of initial osculating elements. A second optimization for a set of virtual ΔV vectors can correct for spatial orbit discontinuities at end-cycles. However, this increases the consecutive repeat cycle space errors as seen in the two distinct spatial error evolution tracks in 2.10 and 2.11. Furthermore, in alignment with simple analytical models, the continuous near-equatorial reference orbit via the virtual maneuvers’ technique was more “delicate” to achieve, and required a much longer convergence time. The introduction of the impulsive ΔV , while resolving end-cycle position discontinuity, violated one of the orbit requirements given in this study, which is to generate a smooth and continuous trajectory in the sense of both positions and velocity. Henceforth, an alternative orbit design scheme will be presented in the next section.

3. New Approach for Reference Orbit Design

In this section, the alternative orbit design methodology requires only one optimization step, replacing the optimization of the virtual ΔV with a simple averaging filter mechanism. This also addresses continuity in all state vectors (position, velocity, acceleration etc.), and it is relevant to near-equatorial orbits where convergence requires a much more careful selection of optimization parameters than if using previous methods.

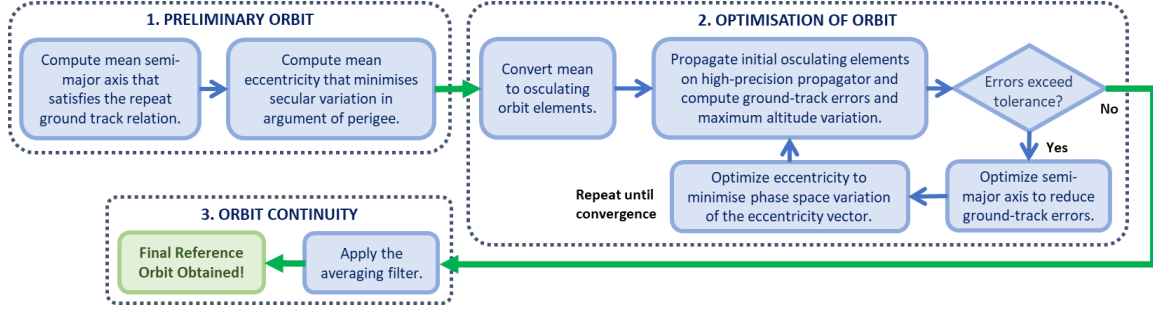


Figure 3.1 – New orbit design scheme using an averaging filter for orbit state continuity.

The averaging filter will be described using the 3.9 day near-equatorial FRO as an example, which has a repeat cycle period equal to 59 nodal periods. The averaged ECEF reference trajectory states $\vec{R}_{Avg}(t)$, $\vec{V}_{Avg}(t)$ are obtained as a simple linear combination of the forward and backward propagated trajectories.

$$\vec{R}_{Avg}(t) = \left(1 - \frac{t}{T_{repeat}}\right) \vec{R}_{prop}(t) + \left(\frac{t}{T_{repeat}}\right) \vec{R}_{prop}(t - T_{repeat}) \quad (16a)$$

$$\vec{V}_{Avg}(t) = \left(1 - \frac{t}{T_{repeat}}\right) \vec{V}_{prop}(t) + \left(\frac{t}{T_{repeat}}\right) \vec{V}_{prop}(t - T_{repeat}) \quad (16b)$$

For $0 \leq t \leq T_{repeat}$, where t is the time elapsed since the beginning of the repeat cycle, T_{repeat} is the repeat cycle period (of 59 orbits in this case), and $\vec{R}_{prop}(t)$, $\vec{V}_{prop}(t)$ are the position and velocity trajectories obtained by numerically propagating the state vectors at the beginning of the repeat cycle. This trivially satisfies the continuity constraints $\vec{R}_{Ref}(0) = \vec{R}_{Ref}(T_{repeat})$ and $\vec{V}_{Ref}(0) = \vec{V}_{Ref}(T_{repeat})$. A simplified visualization of the filter's effects on a toy trajectory is illustrated in Figure 3.2A and 3.2B.

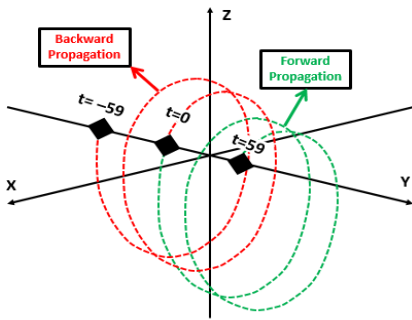


Figure 3.2A – Visualization of trajectory without filtering, showing the discontinuity between the initial and final epochs of each repeat cycle.

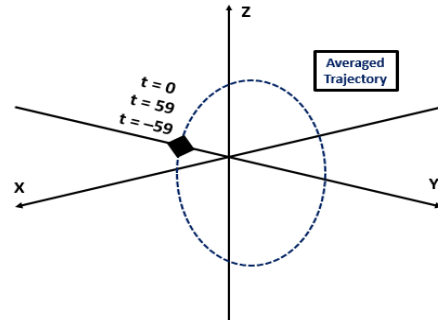


Figure 3.2B – Visualization of trajectory after filtering. No discontinuity is observed between initial and final epochs of each repeat cycle.

To verify that the filter has been successfully applied, there would be two notable observations. First, the state error between the filtered orbit and non-filtered backward propagated orbit decreases as time increases, ending with a zero-error at $t = 0$ (Figure 3.3A). Second, the state error between the filtered orbit and non-filtered forward propagated orbit increases as time increases, starting with a zero error at $t = 0$ (Figure 3.3B). This is because of the linearly increasing and decreasing weights applied on the backward and forward propagated orbits respectively. The same effect applies to both position and velocity (Figure 3.4A/B).

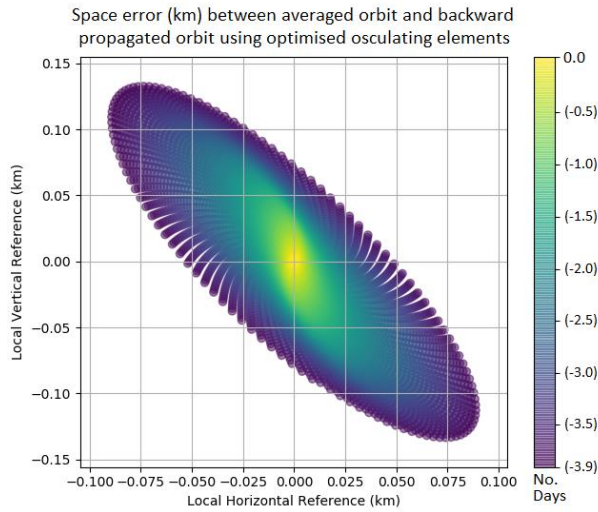


Figure 3.3A – Averaged vs Non-Averaged Spatial Error for the backward propagated orbit

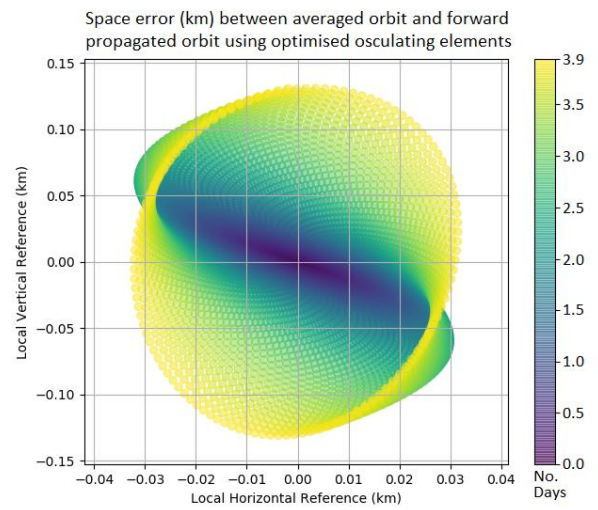


Figure 3.3B – Averaged vs Non-Averaged Spatial Error for the forward propagated orbit

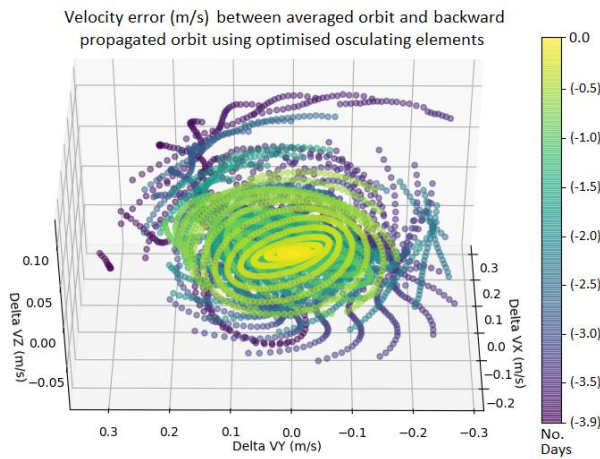


Figure 3.4A – Averaged vs Non-Averaged Velocity Errors in Hill Frame for backward propagated orbit

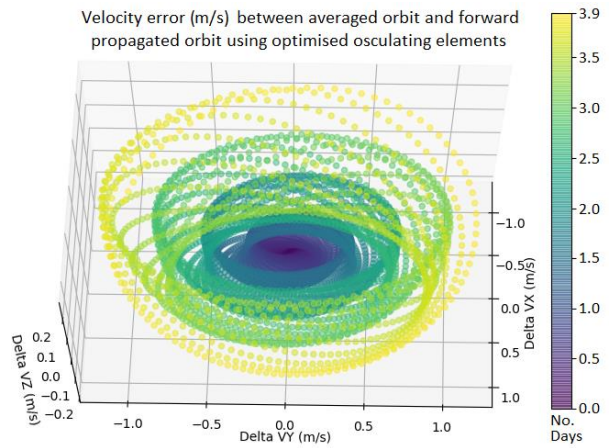


Figure 3.4B – Averaged vs Non-Averaged Velocity Errors in Hill Frame for forward propagated orbit

As a result, the final averaged trajectory contains no end-cycle state discontinuities. In other words, the state at initial and final epochs of the averaged trajectory are the same for both position and velocity. One critical step was that the trajectory provided to the averaging filter was already optimized, such that the FRO end-cycle discontinuities were already an order of magnitude smaller than the tolerance band before filtering (see Table

2.3A). Thus, applying the filter on the optimized orbit did very little to incur orbit modelling errors, and the use of the averaging filter is said to have had minimal impact on the realism of the final reference trajectory. To be certain of this claim, an orbit maintenance control simulation will be performed in the next segment, with a comparison made between the two continuous reference orbits: the averaged reference trajectory and the reference trajectory with virtual maneuvers, as the control target.

4. Validation of the New Reference Orbit Design

In this section, a simulated orbit control strategy is performed. This section aims to validate that the averaged reference trajectory serves as a realistic reference; if it was not, then modelling errors are expected to accumulate over time and one would observe an increasing amount of ΔV needed to compensate for such errors. For the purpose of comparison, this section will also assess the reference trajectory with virtual maneuvers too. The orbit in the “actual” trajectory simulated in this section will inherit the same orbit elements as the near-equatorial trajectory in Table 2.1, using a 500kg micro-satellite with a mean drag area of 20m². The propagator, as before, uses the 70x70 EGM96 geopotential model, with sun and moon perturbations, except with the addition of a Jacchia-Roberts 1976 atmospheric density model and a spherical radiation pressure model.

To give an overview of frozen repeat orbit maintenance, the orbit control of the “actual” trajectory must address atmospheric drag and radiation pressure. Such decay forces decrease the semi-major axis (and altitude) of the satellite over time, thus decreasing the nodal period and introducing an eastward drift in the ground track. With orbit raising, the semi-major axis (and altitude) rises, causing an increase in nodal period, and induces a westward ground track shift instead. The balance between maintaining altitude, as well as westward and eastward drift, is the central focus of the orbit maintenance as described in Figure 4.1. Additionally, in the case of a frozen repeat orbit, the thrusters should be fired only in perigee-apogee pairs in order to minimize any variation to the already-established long-periodic frozen eccentricity. This is best visually described by having the actual satellite maintain a relative orbit about a twin “ghost” satellite as a control target that follows the drag-free reference trajectory as per Low *et al* [12] through an orbital torus with an inner-radius of some tolerance. The reference trajectory goes into the plane of the plot, through the origin in Figure 4.1, where E_R describes the radial (local vertical) spatial error, and E_C describes the cross-track (local horizontal) spatial error.

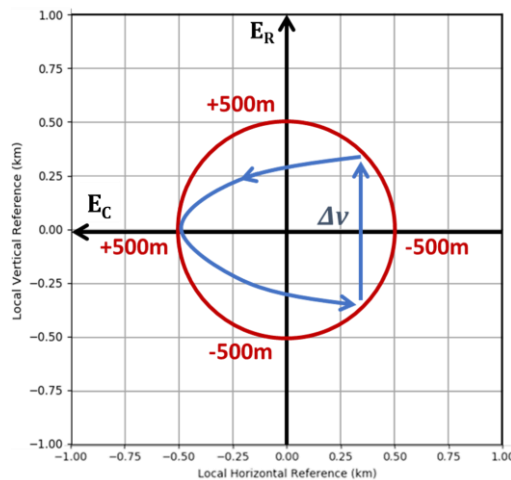


Figure 4.1 – Illustration of the ideal case for spatial error evolution in ECEF

In either case of using the reference with averaging or the reference with virtual maneuvers, the orbit torus tolerance was set to 500m. The control response is triggered when the “actual” spacecraft, with drag forces, crosses the tolerance; almost always in the radial direction rather than the cross track, as observed in Figures 4.4A and 4.4B. When that happens, an impulsive ΔV will be applied to the spacecraft to re-target it back to 500m above the reference altitude (plot origin). Thus, for the purpose of the control simulation, in order to know when to trigger the maintenance maneuver, the space errors must be described. From Kahle *et al* [8], the radial space error E_R and cross-track space error E_C can be described in (17) and (18) respectively as functions of the argument of latitude u , the radial distance R , and the differences between the “actual” and reference trajectories’ osculating elements (semi-major axis difference Δa , inclination difference Δi , right angle of ascending node difference $\Delta \Omega$ and differences in the eccentricity vector components Δe), at some instant in time of sampling.

$$E_R = \Delta a + R(\Delta e_x \cos(u) + \Delta e_y \sin(u)) \quad (17)$$

$$E_C = [R \cdot \Delta \Omega \cdot \sin(i + \alpha) \cos(u)] + [R \cdot \Delta i \sin(u)] \quad (18)$$

In total, 100-days of simulated maintenance was performed, with state vectors sampled at 1-hour intervals. The maneuvers were computed using STK Astrogator’s automatic target sequence, via Broyden’s method as the choice of numerical solver, in similar fashion to how the virtual maneuvers were computed in Figures 2.7 and 2.8 earlier. As a result, Figures 4.4A and 4.4B show the local spatial error plot of the simulated actual orbit, against both the references with averaging and virtual maneuvers, respectively. Figures 4.5A and 4.5B show the spatial error norm evolution across time, using the references with averaging and virtual maneuvers, respectively. Table 4.1 outlines the statistics of the error computations.

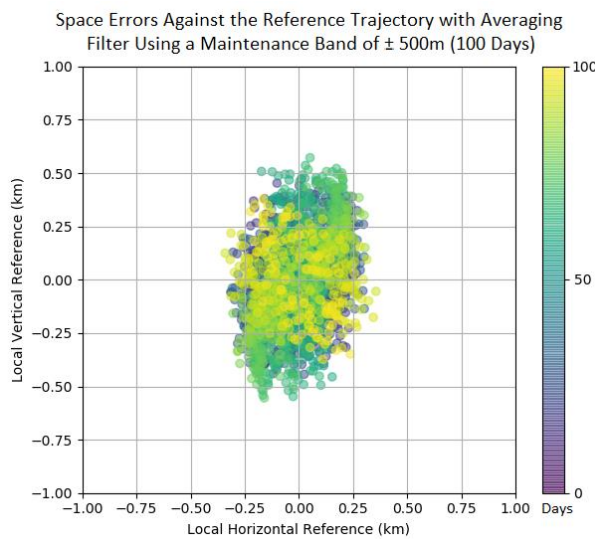


Figure 4.4A – Spatial errors against reference with averaging, sampled every hour for 100 days.

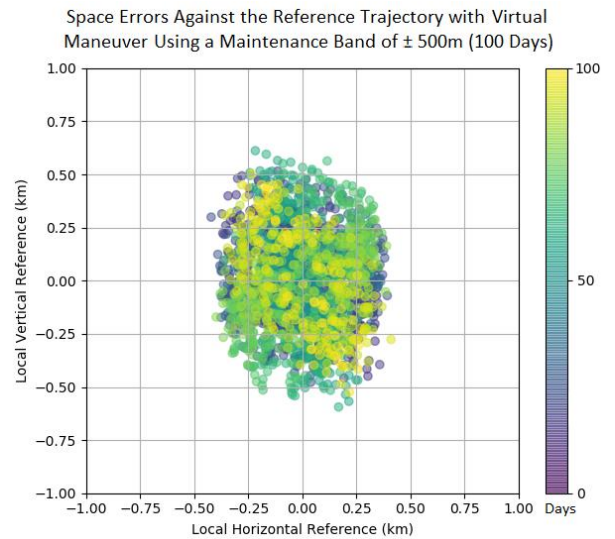


Figure 4.4B – Spatial errors against reference with virtual maneuvers, sampled every hour for 100 days.

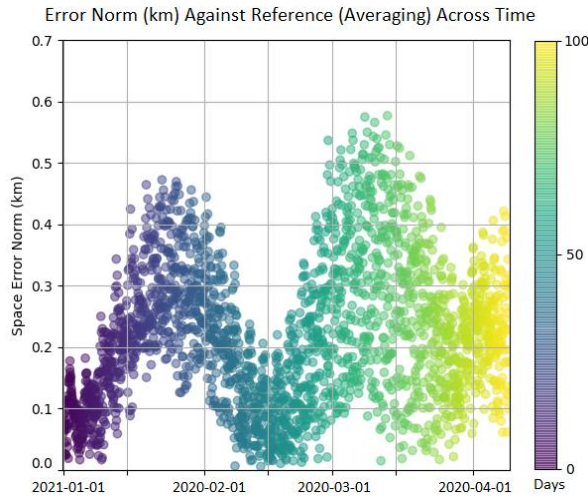


Figure 4.5A – Error norm against reference with averaging, sampled every hour for 100 days.

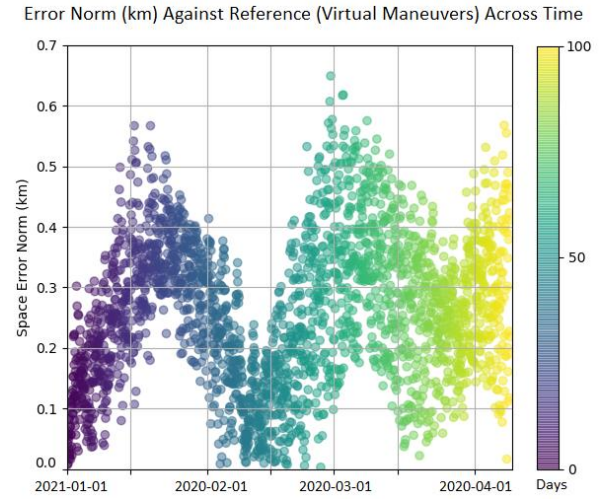


Figure 4.5B – Error norm against reference with virtual maneuvers, sampled every hour for 100 days.

Reference Type	Radial Error Mean (m)	Radial Std. Dev. (m)	X-Track Error Mean (m)	X-Track Std. Dev. (m)	Mean Error Norm (m)
Average-Filtered	6.602	207.014	-1.027	132.512	216.563
Virtual Maneuver	7.430	222.543	-0.041	178.863	258.437

Table 4.1 – Spatial error mean and variance statistics

In summary, the orbit control simulation achieved a mean spatial error characteristically similar to what was achieved for previous polar FRO missions, and was also validated against the method of virtual maneuvers. In both cases, the total consumed ΔV was almost exactly the same: about 5m/s for 100 days of maintenance, and the evolution of their spatial errors are consistent. This means that orbit maintenance about an average-filtered reference would work just as well as a reference generated using virtual maneuvers, as per TerraSAR-X. Interestingly, the average filtered trajectory has noticeable 26% decrease in cross-track error standard deviation as compared to the virtual maneuvered trajectory, despite both cases having employed the same control law but on a different reference target as illustrated in Figure 4.1 earlier.

5. Conclusion

In conclusion, this study was able to apply and validate past techniques used for polar orbit design onto a near-equatorial, circular, frozen, repeat ground track, low Earth orbit as a reference trajectory. Such orbits have not been flown yet at low inclinations, and thus this is a first-time validation. In order to minimize the orbit design time and also ensure orbit continuity with respect to all state vectors (position, velocity, acceleration etc.), an averaging filter technique was developed, and evaluated to be a feasible reference trajectory through an orbit maintenance simulation. The authors sincerely hope that this study would serve as a useful evaluation of past frozen repeat orbit design techniques for near-equatorial orbits, while also proposing a more efficient numerical recipe for generating a smooth and continuous reference orbit trajectory at low inclinations.

6. References

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